as the time derivative of m^2, and $(2m\mathbf{v}) \cdot d(m\mathbf{v})/dt$ is the time derivative of $(mv)^2$. So, Eq. (3.15) is the same as

$$c^2 \frac{d(m^2)}{dt} = \frac{d(m^2 v^2)}{dt}. \tag{3.16}$$

If the derivatives of two quantities are equal, the quantities themselves differ at most by a constant, say C. This permits us to write

$$m^2 c^2 = m^2 v^2 + C. \tag{3.17}$$

We need to define the constant C more explicitly. Since Eq. (3.17) must be true for all velocities, we can choose a special case where $v = 0$, and say that in this case the mass is m_0. Substituting these values into Eq. (3.17) gives

$$m_0^2 c^2 = 0 + C\,.$$

We can now use this value of C in Eq. (3.17), which becomes

$$m^2 c^2 = m^2 v^2 + m_0^2 c^2. \tag{3.18}$$

Dividing by c^2 and rearranging terms gives

$$m^2(1 - v^2/c^2) = m_0^2,$$

from which we get

$$m = m_0/\sqrt{1 - v^2/c^2}\,. \tag{3.19}$$

This is the formula (3.1), and is exactly what is necessary for the agreement between mass and energy in Eq. (3.12).

Ordinarily these energy changes represent extremely slight changes in mass, because most of the time we cannot generate much energy from a given amount of material; but in an atomic bomb of explosive energy equivalent to 20 kilotons of TNT, for example, it can be shown that the dirt after the explosion is lighter by 1 gram than the initial mass of the reacting material, because of the energy that was released, i.e., the released energy had a mass of 1 gram, according to the relationship $\Delta E = \Delta(mc^2)$. This theory of equivalence of mass and energy has been beautifully verified by ex-